

StudyFlow AI – Platform Operations & Access Guide

A complete reference guide for engineers, administrators, educators, and students. Covers codebase acquisition, local setup, admin console governance, and all section workflows.

FRAMEWORK	BACKEND / AUTH	AI ENGINE	DOCUMENT DATE
TanStack Start v1 (React 19)	Lovable Cloud / Supabase RLS	TF-IDF & Cosine Similarity	September 2026 (v1.4)

1. How to Get & Export the Codebase

StudyFlow AI is built with modern web standards and full portability. You can export, sync, and run the complete project locally on any developer machine.

Option A: Connect & Push to GitHub (Recommended)

1. Inside the Lovable project editor interface, locate the **GitHub** icon at the top right of the navigation bar.
2. Click **Connect to GitHub** (or navigate to *Settings* → *GitHub Sync*).
3. Authorize your GitHub account and select an existing repository or create a new private/public repository.
4. Lovable automatically commits and pushes all branches, source files, migrations, and assets to your repository with continuous bi-directional synchronization.
5. Once connected, clone the repository to your workstation:

```
git clone https://github.com/your-username/studyflow-ai.git
cd studyflow-ai
```

Option B: Local Development Setup & Execution

The application runs on TanStack Start v1 with Vite 7 and works smoothly with Bun or Node.js (v20+):

```
# 1. Install project dependencies
bun install # or: npm install

# 2. Verify environment configuration (.env)
# VITE_SUPABASE_URL=https://zcselumficemtnqgioq.supabase.co
# VITE_SUPABASE_PUBLISHABLE_KEY=sb_publishable_...

# 3. Launch local dev server
bun run dev # or: npm run dev -> Runs at http://localhost:8080

# 4. Execute automated test suite
bunx vitest run
```

Project Structure Quick Map

- `src/routes/` : TanStack file-based routing (public routes, `_authenticated/` tree, and `admin.*` routes).
- `src/lib/` : Core business logic, TF-IDF recommendation engine (`recommendation.ts`), and server RPC functions.
- `drizzle/migrations/` : Version-controlled database schema migrations and RLS security policies.
- `src/components/study/` : Design-system components, navigation app shell, and role guards.

2. How to Open & Operate the Admin Panel

The Admin Panel is a dedicated governance cockpit with strict Row-Level Security (RLS) enforcement. Only authenticated accounts possessing the `admin` role in the `public.user_roles` table can view or execute admin actions.

Admin Credentials

Primary provisioned administrator account:

Email: `vikramkumarsingh.unxt@gmail.com`

Role: `admin` (verified in `public.user_roles`)

Sign in via `/login`, then navigate to `/admin/dashboard` or click the Admin shield in the top header.

HIGH PRIVILEGE

Security Definer Guard

All administrative routes are protected by `<AdminGuard>`. If an unauthenticated or non-admin user accesses any `/admin/*` URL, the client immediately blocks rendering and displays an unauthorized access shield with a direct redirect to safe pages.

RBAC

Admin Panel Routes & Operational Capabilities

Admin Route	Interface Name	Core Actions & Governance Capabilities
<code>/admin/dashboard</code>	Admin Overview	Executive summary showing pending approvals, active materials, registered learners, verified educators, and moderation shortcuts.
<code>/admin/materials</code>	Approval Queue	Review community/teacher submissions. Features in-app content preview (PDF/video/text), approve to publish to catalog, or reject with feedback reason.
<code>/admin/materials/new</code>	Create Material	Directly publish official institutional resources with file uploads, tags, subject selection, and author attribution.
<code>/admin/materials/\$id/edit</code>	Edit Resource	Update existing resource title, content URL, difficulty level, subject classification, and linked verified educator.
<code>/admin/subjects</code>	Subject Taxonomies	Create, rename, and manage academic subjects and curriculum categories across the platform.
<code>/admin/reports</code>	Reports & Feedback	Monitor student question and comment feedback submitted on study materials, track rating trends, and download full CSV exports.
<code>/admin/authors</code>	Verified Authors	Register educator profiles (name, bio, academic institution, avatar URL) and link them to published study resources.
<code>/admin/educators</code>	Educator Queue & Edits	Review teacher onboarding applications, approve/link teacher accounts to author profiles, and review/apply educator material correction requests.

How to Grant Admin Privileges via Database SQL

```
-- Grant admin role to a user account
INSERT INTO public.user_roles (user_id, role)
SELECT id, 'admin':public.app_role
FROM auth.users
WHERE email = 'target-user@example.com'
ON CONFLICT (user_id, role) DO NOTHING;
```

⚠ Security Architecture: Separation of Roles

User roles are strictly isolated in `public.user_roles` and verified through the `public.has_role(auth.uid(), 'admin')` security-definer function. Roles are never stored on client-accessible profiles to prevent privilege escalation.

3. Instructions to Access Student & Learning Pages

Student workflows are engineered for fast, frictionless onboarding and cold-start-proof intelligent recommendations.

Cold-Start Student Onboarding
/ONBOARDING

When a student signs in for the first time, navigation is intercepted to guide them through:

- Selecting primary academic goal (Exam prep, deep mastery, quick review).
- Choosing target subject (Computer Science, Machine Learning, etc.).
- Entering specific starting topics to bootstrap TF-IDF vectors instantly.

Explainable Recommendations
/RECOMMENDATIONS

StudyFlow AI shows transparent reasoning for every suggested material:

- Cosine similarity score badge (e.g. 94% match).
- Exact overlapping topic tags that triggered the match.
- Subject alignment indicator based on historical viewing and ratings.

Student Platform Directory

Page Path	Access Rule	Description & User Features
/register & /login	PUBLIC	Email and password authentication. Features toggle links to Educator Sign-up and automated session persistence.
/onboarding	STUDENT	Interactive cold-start preferences setup. Teachers and Admins automatically bypass this step.
/dashboard	STUDENT	Personalized student hub with quick stats (materials viewed, ratings submitted, study hours), latest recommendations, and active study groups.
/search	STUDENT	Searchable public catalog. Filter by academic subject, resource type (notes, video, cheat sheet, lab), and difficulty rating.
/materials/\$id	STUDENT	Full material detail page: view attached PDF/content, submit 1–5 star ratings with written reviews, and ask questions or post comments in the feedback section.
/recommendations	STUDENT	AI-curated recommendations based on student's TF-IDF preference vector with matching explanation tags.
/study-groups	STUDENT	Collaborative study rooms: join subject groups, study teacher-shared resources, check off completed notes, and monitor group-wide progress counters.
/submit	STUDENT	Upload peer-study materials (title, summary, subject, tags, and document attachment) into the moderation queue.
/submissions	STUDENT	Track personal submission history with status badges (Pending Review, Approved, Rejected) and reviewer feedback notes.
/history & /profile	STUDENT	Review your learning history timeline and adjust your target academic goals and subject preferences at any time.

☐ Seeded Demo Student Accounts (Instant Testing)

Alex Chen:

demo.alex+studyflow@example.com

Pass: StudyFlow!2026

Maya Patel:

demo.maya+studyflow@example.com

Pass: StudyFlow!2026

Sam Rivera:

demo.sam+studyflow@example.com

Pass: StudyFlow!2026

4. Instructions for Educator Portal & Teacher Workflows

The Educator Portal provides university professors and certified teachers with dedicated tools to publish courseware, track learner engagement, and maintain material accuracy.

Educator Onboarding & Application

/EDUCATOR-SIGNUP

- Navigate to `/educator-signup` (also linked from landing page and register page).
- Enter name, educator email, password, and select subject expertise chips (e.g., Computer Science, Machine Learning).
- Upon account creation, the teacher is automatically guided to `/educator/apply` to submit institution and professional credentials.
- An administrator reviews and approves the application at `/admin/educators`, granting the `teacher` role and linking the verified author profile.

Educator Dashboard Features

/EDUCATOR

- Approved Material Metrics:** Real-time view counts and aggregate engagement across all authored materials.
- Most-Viewed Ranking:** Highlights your highest-impact study guides and labs.
- Study Group Adoption:** Shows which active study groups are utilizing your shared resources.
- Submit Material:** Fast-track submission directly attributed to your verified educator profile.
- Request Material Edits:** Submit correction requests to admins with review tracking (Open, Applied, Declined).

👤 Active Demo Educator Account

Educator: Priya Raman (Machine Learning Faculty)

Email: `demo.teacher+studyflow@example.com` | **Password:** `StudyFlow!2026`

Linked to published resources including "Gradient descent intuition lab (PyTorch)" and "Search and heuristics in classical AI".

5. Complete Master Routing & Credentials Summary

URL Path	Role Required	Key User / Credential	Primary Purpose
<code>/</code>	PUBLIC	Anyone	Landing page with feature showcase, quick login, and educator sign-up links.
<code>/login</code> & <code>/register</code>	PUBLIC	Student or Any User	Account creation and authentication entry points.
<code>/educator-signup</code>	PUBLIC	New Teachers	Educator registration with subject expertise selection.
<code>/educator/apply</code>	AUTHENTICATED	Pending Teachers	Teacher credential submission for admin review.
<code>/educator</code>	TEACHER	<code>demo.teacher+studyflow@...</code>	Teacher dashboard: views, metrics, material submissions, edit requests.
<code>/dashboard</code>	STUDENT	<code>demo.alex+studyflow@...</code>	Student learning dashboard with recommendations and stats.
<code>/search</code>	STUDENT	Any authenticated learner	Full resource catalog search with multi-faceted filtering.
<code>/study-groups</code>	STUDENT	Any authenticated learner	Study group collaboration, shared resources, and progress tracking.
<code>/submissions</code>	STUDENT	Any authenticated learner	Community peer submissions and review status tracker.
<code>/admin/dashboard</code>	ADMIN	<code>vikramkumarsingh.unxt@...</code>	Governance center and administrative metrics.
<code>/admin/materials</code>	ADMIN	<code>vikramkumarsingh.unxt@...</code>	Approval queue: preview content, approve, or reject submissions.
<code>/admin/educators</code>	ADMIN	<code>vikramkumarsingh.unxt@...</code>	Review teacher applications and apply material edit requests.
<code>/admin/reports</code>	ADMIN	<code>vikramkumarsingh.unxt@...</code>	User feedback management and CSV data exports.